Arihan Iyer

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Education

University of Michigan — Ann Arbor, MI
M.S. in Computer Science & Engineering — *Aug 2026 – May 2028*
- Coursework: Advanced Scalable Systems, Natural Language Processing, Human-AI Interaction

University of Mumbai — Mumbai, India
B.E. in Information Technology, Honours in AI & ML; GPA: 3.90/4.00 — *Aug 2021 – May 2025*
- Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Databases, Computer Networks, Machine Learning, Information Retrieval

Experience

AI Solutions Specialist — Oct 2026 – Present
Business+Tech, Ross School of Business, University of Michigan — *Ann Arbor, MI*
- Guide **12 student teams** developing AI solutions for a real-estate client; advise on LLM use cases, evaluation-set design, and technical architecture during the Fall 2026 AI Sprint.

Founding Software Engineer (Innovation Architect) — Jun 2025 – Jul 2026
CreatED — *Hybrid, Mumbai*
- As the first technical hire, led engineering delivery of **30+ internal and client software/AI projects** from requirements through deployment.
- Automated weekly progress reporting for **100+ students and 50+ mentors** through COSMIC, building PostgreSQL data extraction and scheduled Node.js reports that saved **∼30 administrative hours monthly** and synchronized learner data with Sync.
- Automated Google Meet scheduling, homework distribution, and homework-review routing across **100+ WhatsApp groups**; built webhook-driven workflows and Redis-backed queues to process **400+ messages weekly**.
- Built CEDAR AI, a research-writing assistant used by **25+ researchers to draft and format 80+ manuscripts**; implemented PDF ingestion, retrieval-augmented generation over ChromaDB, and structured JSON outputs, with Dockerized services automatically deployed through GitHub Actions.

Chief Technology Officer, Healthcare (Part-time) — Aug 2024 – May 2025
100 Girls in Generative AI — *Mumbai, India*
- Led development of a Hindi-English nutrition chatbot to make public-health guidance more accessible to rural mothers, grounding responses in curated health resources through RAG.
- Enabled offline use on **8 GB RAM laboratory machines** by deploying a 4-bit GGUF-quantized Llama 3 8B with llama.cpp for GPU-assisted inference, reaching **∼10 tokens/sec** without cloud dependency.

Software Development Engineer Intern (Part-time) — Apr 2024 – Feb 2025
LearnMigo — *Hybrid, Mumbai*
- Built configurable avatar chatbots for a subject-aware AI teaching assistant, piloted by **5 educators reaching 200+ students**; used RAG and subject-specific prompts to tailor responses to learning objectives.
- Automated drafting of lesson content and presentations for teachers by building LessonAI with reusable prompt templates and OpenAI Structured Outputs (JSON Schema), validated through automated pytest suites.

Selected Projects

Countersign | *Jac, MCP, Stripe API, LLM-as-judge* — 2026
- Built a payment-guard server for autonomous AI agents making test-mode purchases, enforcing deterministic budget, price-consistency, and URL-safety checks alongside LLM-based verification of merchant legitimacy and purchase intent.
- Evaluated the payment guard across **40+ benign and adversarial purchase scenarios**, including prompt-injection attempts; approved **95% of 20+ legitimate purchases**. **Won the IBM Track and Best Use of Jaclang at JacHacks 2026**, the only team among approximately 50 to win two tracks.

Reasoning Canvas | *React, TypeScript, FastAPI, pgvector, Gemini* — 2026
- Helped high-school researchers critically examine scientific claims by building an interactive argument canvas that maps each claim against retrieved supporting and opposing evidence.
- Grounded claims in **100+ scientific papers** through RAG over pgvector, using recursive chunking and text-embedding-3-small embeddings; added Gemini search grounding when retrieved evidence was insufficient.

Crop Suitability Pipeline | *Python, scikit-learn, LightGBM, SHAP* — 2025
- Built a two-stage crop recommender to replace generic planting advice for farmers: it recommends crops from soil-nutrient data, then verifies each recommendation against yield thresholds, using Random Forest feature selection and LightGBM classification on **8,000+ geospatial records**.
- Achieved **0.94 macro F1** on a stratified 20% held-out test set; used SHAP to show which inputs drive each recommendation, so farmers can understand and verify it for their own land.

Technical Skills

Languages: Python, C++, TypeScript, JavaScript, SQL, Jac, Dart
Applied AI: RAG, Agentic Workflows, MCP, Structured Outputs, LLM-as-judge, Edge Inference (llama.cpp)
ML: PyTorch, TensorFlow, scikit-learn, Random Forest, LightGBM, SHAP, Logistic Regression, Collaborative Filtering
Frameworks & Data: FastAPI, Node.js, React, Next.js, PostgreSQL, pgvector, Supabase, ChromaDB, Redis, LangChain
Cloud & Tools: AWS (EC2, S3, Lambda), Docker, GitHub Actions, pytest, Stripe API, Git, Vercel

Awards & Leadership

Judge, MHacks 2026 | Finalist, Smart India Hackathon 2022 | President, Information Technology Students Association (led a 100-person organizing team, 2023–2024)